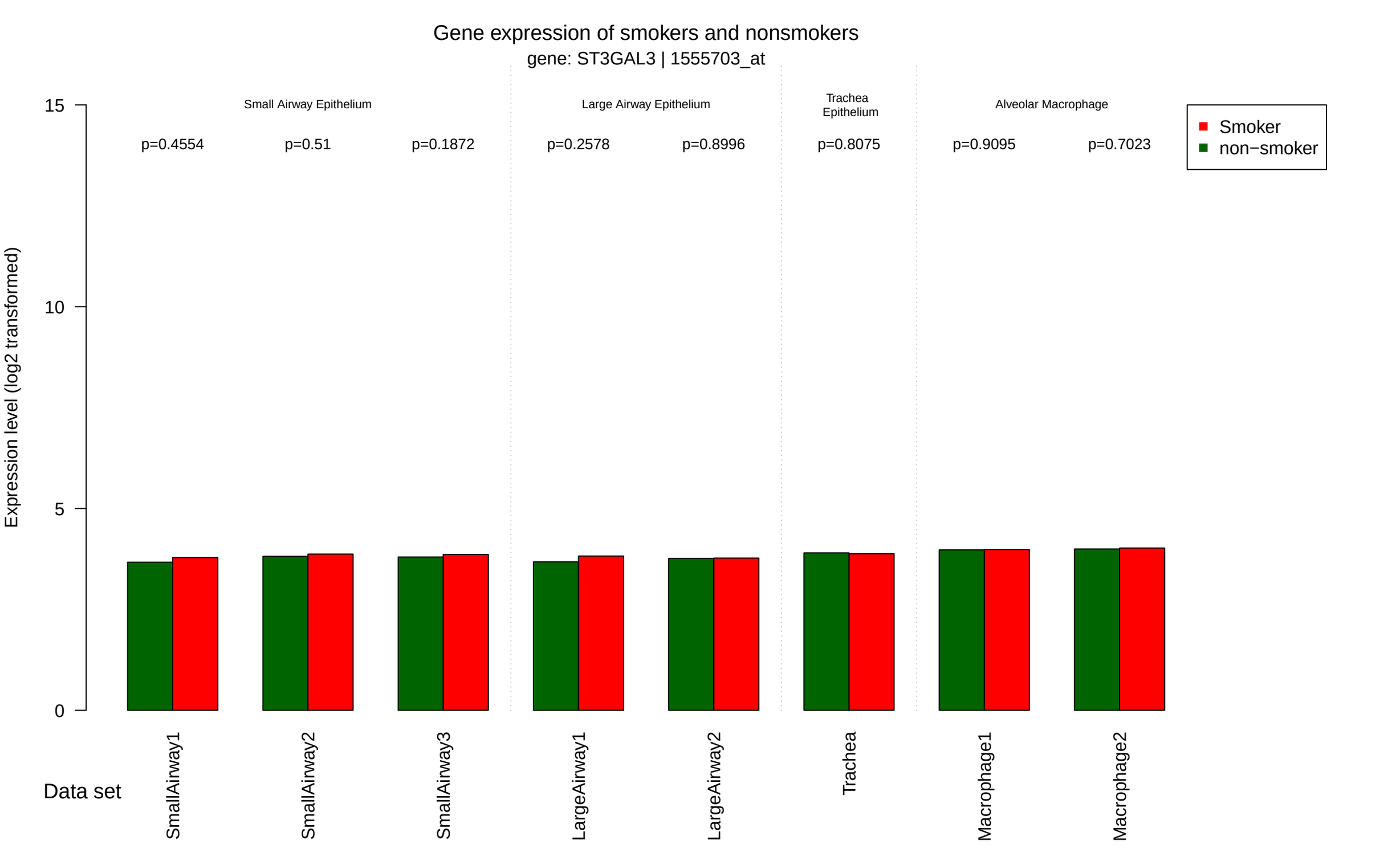
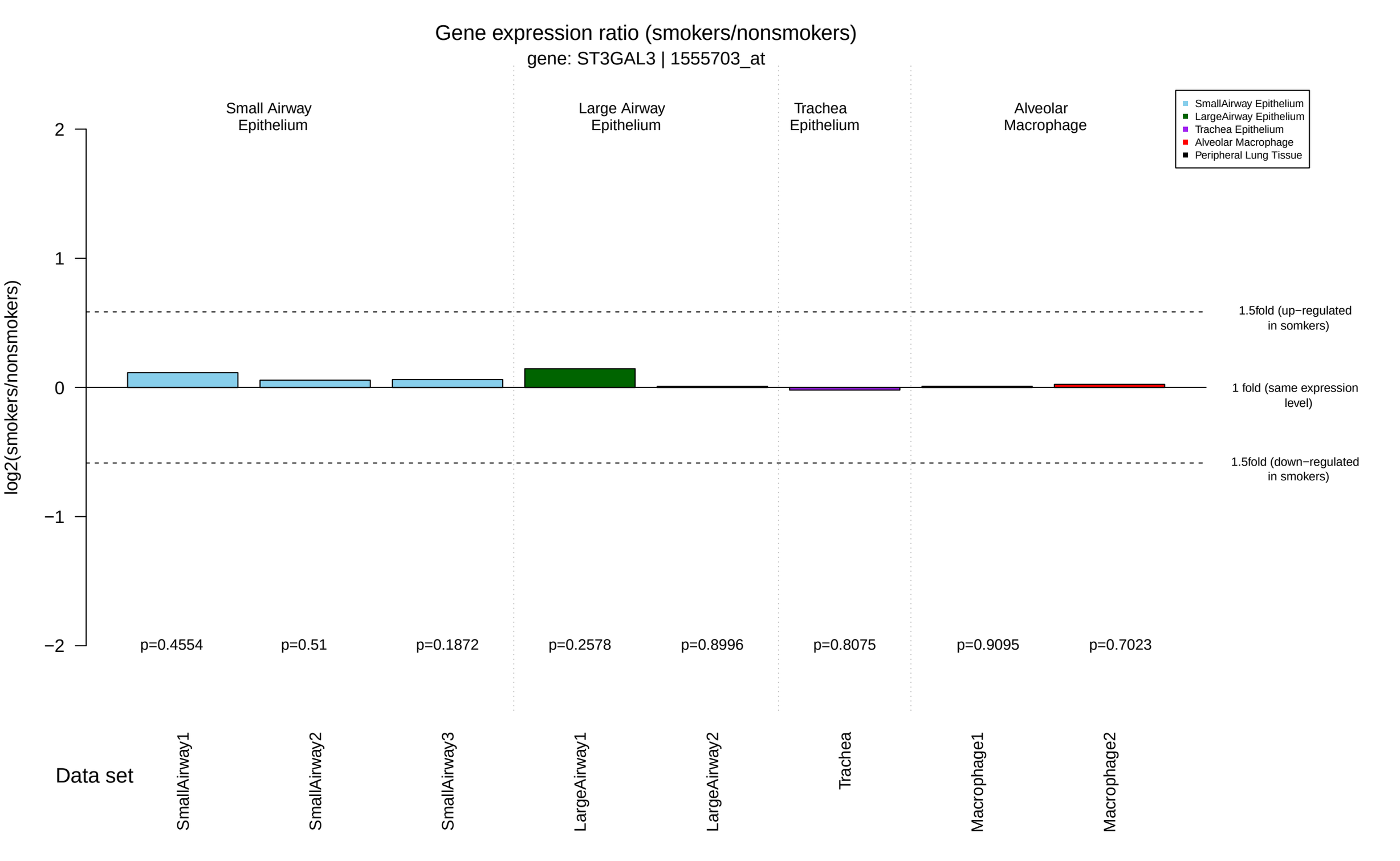
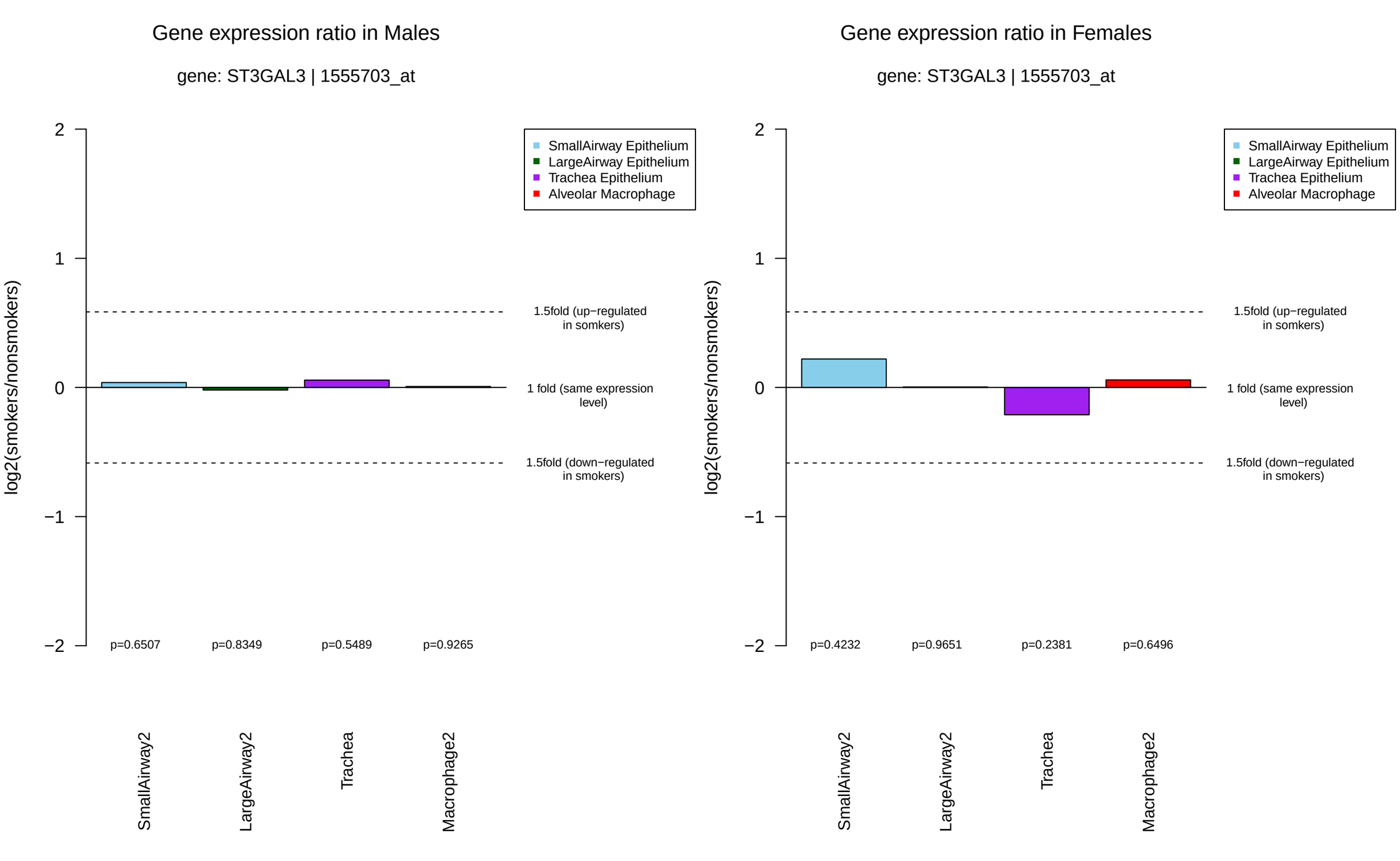
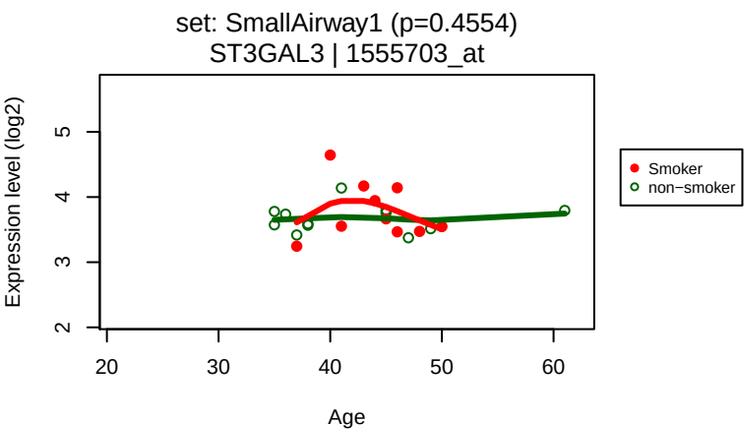
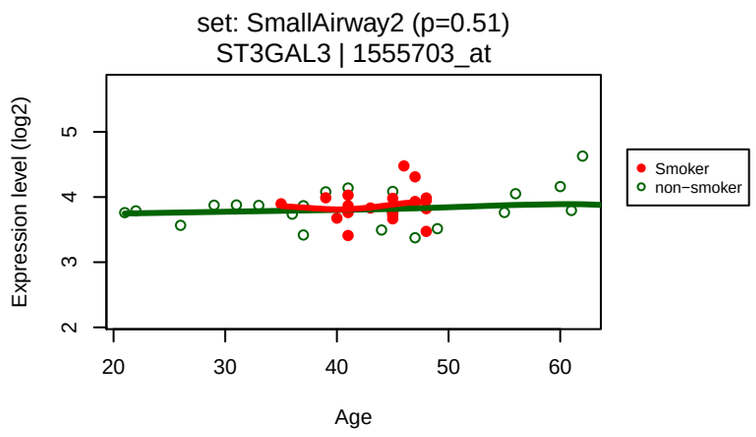
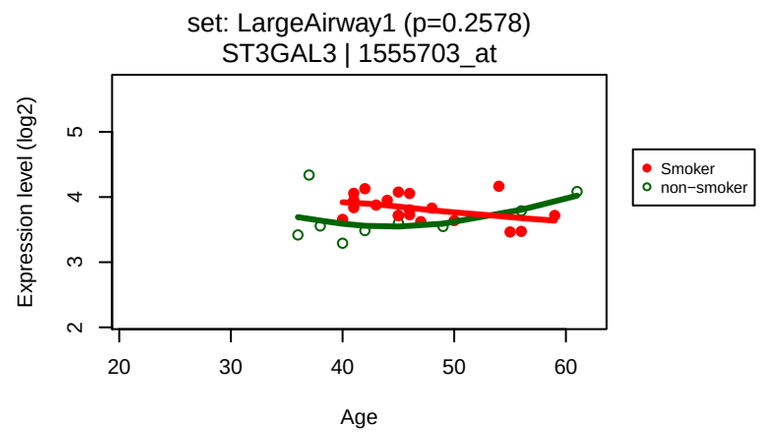

Gene expression of smokers and nonsmokers
gene: ST3GAL3 | 1555703_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4554
p=0.51
p=0.1872
p=0.2578
p=0.8996
p=0.8075
p=0.9095
p=0.7023
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

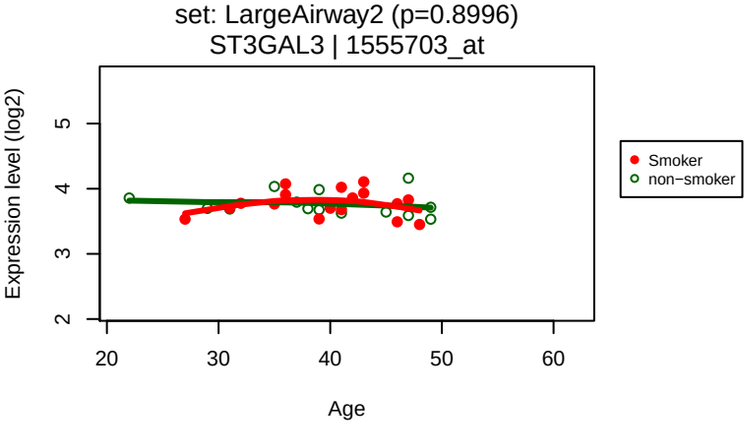

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL3 | 1555703_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4554
p=0.51
p=0.1872
p=0.2578
p=0.8996
p=0.8075
p=0.9095
p=0.7023
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

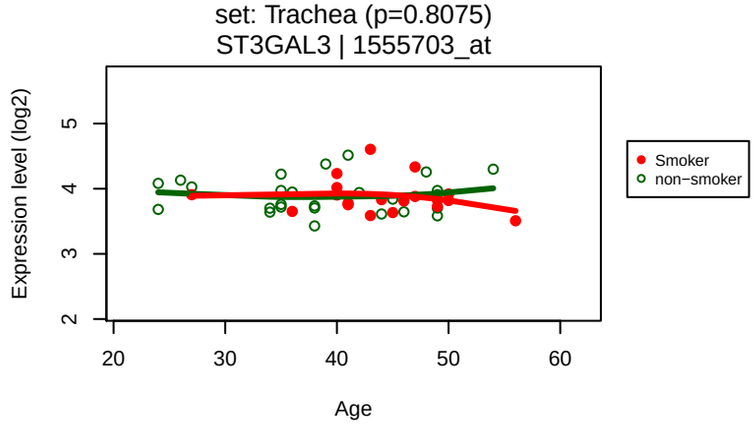

Gene expression ratio in Males
gene: ST3GAL3 | 1555703_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6507
p=0.8349
p=0.5489
p=0.9265
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL3 | 1555703_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4232
p=0.9651
p=0.2381
p=0.6496
SmallAirway2
LargeAirway2
Trachea
Macrophage2

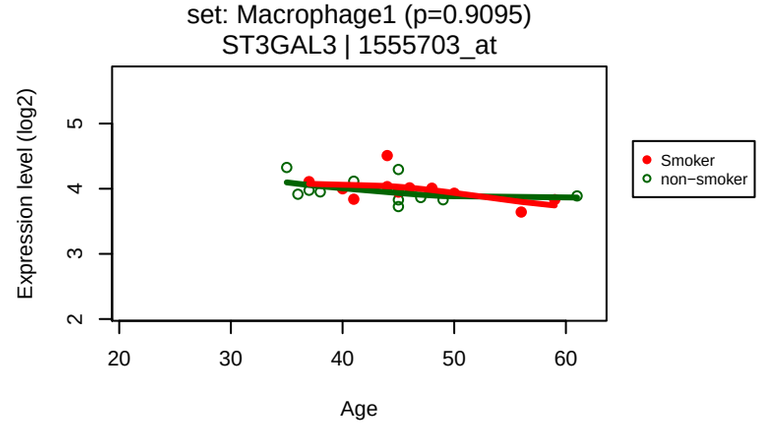

set: SmallAirway1 (p=0.4554)
ST3GAL3 | 1555703_at
Expression level (log2)
Age
Smoker
non−smoker

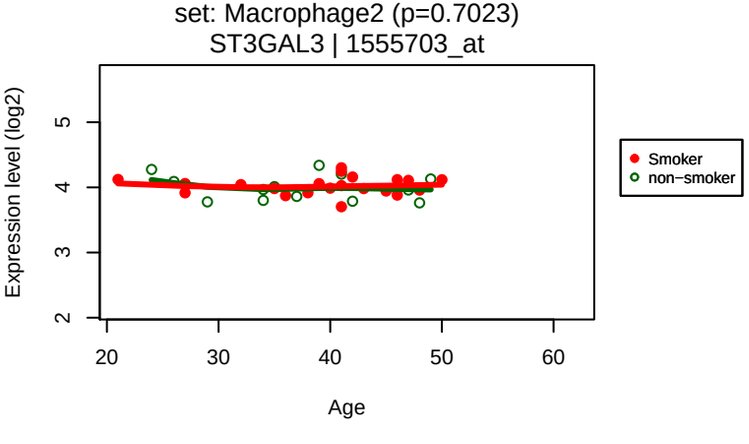

set: SmallAirway2 (p=0.51)
ST3GAL3 | 1555703_at
Expression level (log2)
Age
Smoker
non−smoker

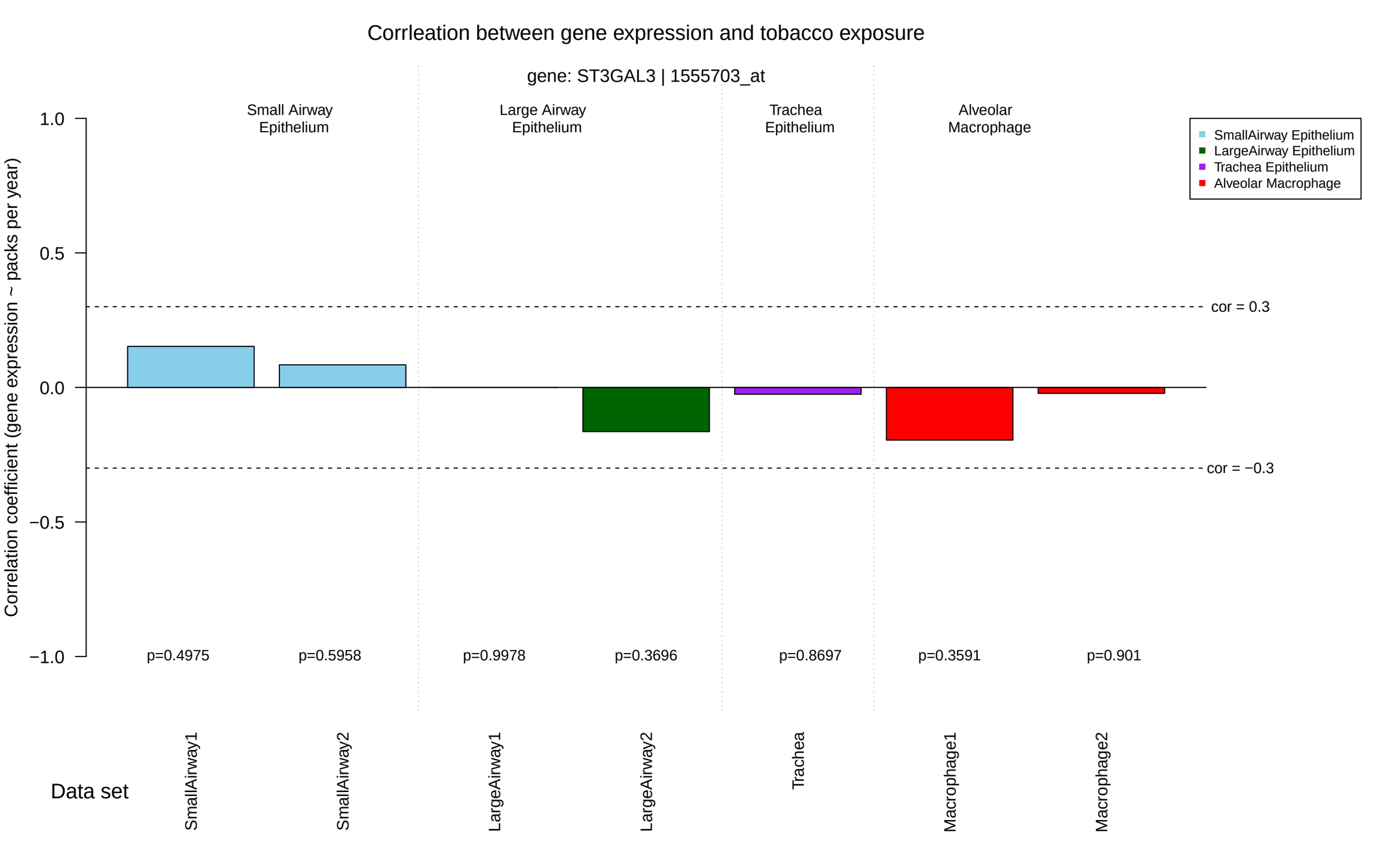

set: LargeAirway1 (p=0.2578)
ST3GAL3 | 1555703_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8996)
ST3GAL3 | 1555703_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8075)
ST3GAL3 | 1555703_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9095)
ST3GAL3 | 1555703_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7023)
ST3GAL3 | 1555703_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL3 | 1555703_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4975
p=0.5958
p=0.9978
p=0.3696
p=0.8697
p=0.3591
p=0.901
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2